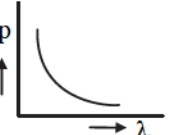
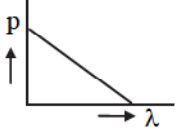
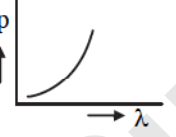
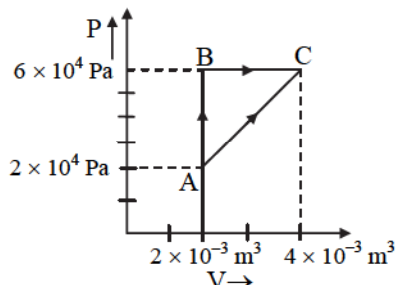


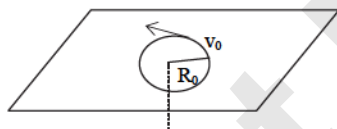
AIPMT - 2015 Question Paper

1. Two particles of masses m_1, m_2 move with initial velocities u_1 and u_2 . On collision, one of the particles get excited to higher level, after absorbing energy ε . If final velocities of particles be v_1 and v_2 then we must have
 - (A) $m_1^2 u_1 + m_2^2 u_2 - \varepsilon = m_1^2 v_1 + m_2^2 v_2$
 - (B) $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \varepsilon$
 - (C) $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 - \varepsilon = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$
 - (D) $m_1^2 u_1^2 + \frac{1}{2} m_2^2 u_2^2 + \varepsilon = \frac{1}{2} m_1^2 v_1^2 + \frac{1}{2} m_2^2 v_2^2$
2. Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength?
 - (A) 
 - (B) 
 - (C) 
 - (D) 
3. If energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be
 - (A) $[E V^{-2} T^{-1}]$
 - (B) $[E V^{-1} T^{-2}]$
 - (C) $[E V^{-2} T^{-2}]$
 - (D) $[E^{-2} V^{-1} T^{-3}]$
4. Across metallic conductor of non-uniform cross section a constant potential difference is applied. The quantity which remains constant along the conductor
 - (A) current density
 - (B) current
 - (C) drift velocity
 - (D) electric field
5. Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the centre. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is
 - (A) -20 cm
 - (B) -25 cm
 - (C) -50 cm
 - (D) 50 cm
6. A ship A is moving westwards with a speed of 10 km h^{-1} and a ship B 100 km south of A, is moving northwards with a speed of 10 km h^{-1} . The time after which the distance between them becomes shortest, is
 - (A) 0 h
 - (B) 5 h
 - (C) $5\sqrt{2} \text{ h}$
 - (D) $10\sqrt{2} \text{ h}$
7. For a parallel beam of monochromatic light of wavelength ' λ ' diffraction is produced by a single slit whose width ' a ' is of the order of the wavelength of light. If ' D ' is the distance of the screen from the slit, the width of the central maxima will be
 - (A) $\frac{2D\lambda}{a}$
 - (B) $\frac{D\lambda}{a}$
 - (C) $\frac{Da}{\lambda}$
 - (D) $\frac{2Da}{\lambda}$
8. A particle of mass m is driven by a machine that delivers a constant power k watts. If the particle starts from rest the force on the particle at time t is
 - (A) $\sqrt{\frac{mk}{2}} t^{\frac{1}{2}}$
 - (B) $\sqrt{mk} t^{\frac{1}{2}}$
 - (C) $\sqrt{2mk} t^{\frac{1}{2}}$
 - (D) $\frac{1}{2} \sqrt{mk} t^{\frac{1}{2}}$
9. Kepler's third law states that square of period of revolution (T) of a planet around the sun, is proportional to third power of average distance r between sun and planet i.e., $T^2 = Kr^3$ here K is constant. If the masses of sun and planet are M and m respectively then as per Newton's law of gravitation force of attraction between them is $F = \frac{GMm}{r^2}$, here G is gravitational constant. The relation between G and K is described as
 - (A) $GK = 4\pi^2$
 - (B) $GK = 4\pi^2$
 - (C) $K = G$
 - (D) $K = \frac{1}{G}$
10. A particle is executing S.H.M along a straight line. Its velocities at distances x_1 and x_2 from the mean position are v_1 and v_2 respectively. Its time period is
 - (A) $2\pi \sqrt{\frac{x_1^2 + x_2^2}{v_1^2 + v_2^2}}$
 - (B) $2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$
 - (C) $2\pi \sqrt{\frac{v_1^2 + v_2^2}{x_1^2 + x_2^2}}$
 - (D) $2\pi \sqrt{\frac{v_1^2 - v_2^2}{x_1^2 - x_2^2}}$
11. Two similar springs P and Q have spring constants K_P and K_Q such that $K_P > K_Q$. They are stretched first by the same amount (case a), then by the same force (case b). The work done by the springs W_P and W_Q are related as, in case (a) and case (b), respectively
 - (A) $W_P = W_Q$; $W_P > W_Q$
 - (B) $W_P = W_Q$; $W_P = W_Q$
 - (C) $W_P > W_Q$; $W_Q > W_P$
 - (D) $W_P < W_Q$; $W_Q < W_P$

12. Figure below shows two paths that may be taken by a gas to go from a state A to a state C. In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be



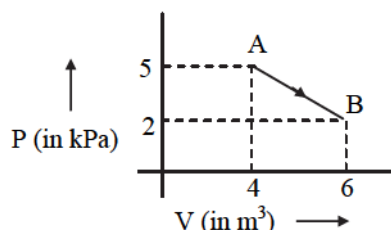
- (A) 380 J (B) 500 J
(C) 460 J (D) 300 J
13. When two displacements represented by $y_1 = a \sin(\omega t)$ and $y_2 = b \cos(\omega t)$ are superimposed the motion is
(A) not a simple harmonic.
(B) simple harmonic with amplitude $\frac{a}{b}$.
(C) simple harmonic with amplitude $\sqrt{a^2 + b^2}$.
(D) simple harmonic with amplitude $\frac{(a+b)}{2}$.
14. A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0 . The mass is attached to a string which passes through a smooth hole in the plane as shown



The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$. The final value of the kinetic energy is

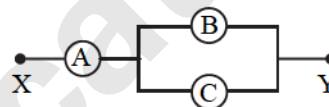
- (A) mv_0^2 (B) $\frac{1}{4}mv_0^2$
(C) $2mv_0^2$ (D) $\frac{1}{2}mv_0^2$
15. If radius of the ${}^{27}_{13}\text{Al}$ nucleus is taken to be R_{Al} then the radius of ${}^{125}_{53}\text{Te}$ nucleus is nearly
(A) $\left(\frac{53}{13}\right)^{1/3} R_{Al}$ (B) $\frac{5}{3} R_{Al}$
(C) $\frac{3}{5} R_{Al}$ (D) $\left(\frac{13}{53}\right)^{1/3} R_{Al}$
16. The electric field in a certain region is acting radially outward and is given by $E = Ar$. A charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by
(A) $4\pi\epsilon_0 Aa^2$ (B) $A\epsilon_0 a^2$
(C) $4\pi\epsilon_0 Aa^3$ (D) $\epsilon_0 Aa^3$

17. One mole of an ideal diatomic gas undergoes a transition from A to B along a path AB as shown in the figure,



The change in internal energy of the gas during the transition is

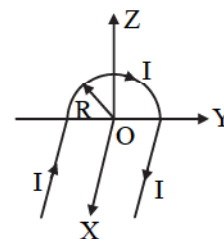
- (A) 20 kJ (B) -20 kJ
(C) 20 J (D) -12 kJ
18. A, B and C are voltmeters of resistance R , $1.5R$ and $3R$ respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are V_A , V_B and V_C respectively, then



- (A) $V_A = V_B = V_C$ (B) $V_A \neq V_B = V_C$
(C) $V_A = V_B \neq V_C$ (D) $V_A \neq V_B \neq V_C$
19. A rod of weight W is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at distance x from A. The normal reaction on A is
(A) $\frac{Wx}{d}$ (B) $\frac{Wd}{x}$
(C) $\frac{W(d-x)}{x}$ (D) $\frac{W(d-x)}{d}$

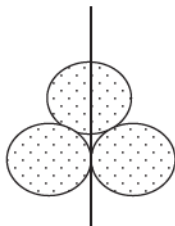
20. A wire carrying current I has the shape as shown in adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semicircular portion of radius R is lying in Y-Z plane. Magnetic field at point O is

- (A) $\vec{B} = \frac{\mu_0 I}{4\pi R} (\pi\hat{i} + 2\hat{k})$
(B) $\vec{B} = -\frac{\mu_0 I}{4\pi R} (\pi\hat{i} - 2\hat{k})$
(C) $\vec{B} = -\frac{\mu_0 I}{4\pi R} (\pi\hat{i} + 2\hat{k})$
(D) $\vec{B} = \frac{\mu_0 I}{4\pi R} (\pi\hat{i} - 2\hat{k})$



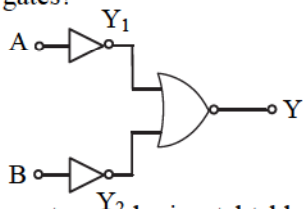
21. Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' which is touching to two shells and passing through diameter of third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is

- (A) $\frac{11}{5} \pi r^2$
 (B) $3 \pi r^2$
 (C) $\frac{16}{5} \pi r^2$
 (D) $4 \pi r^2$



22. Which logic gate is represented by the following combination of logic gates?

- (A) OR
 (B) NAND
 (C) AND
 (D) NOR



23. A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of table and from its other end another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and table is μ_k . When the block A is sliding on the table, the tension in the string is

- (A) $\frac{(m_2 + \mu_k m_1)g}{(m_1 + m_2)}$ (B) $\frac{(m_2 - \mu_k m_1)g}{(m_1 + m_2)}$
 (C) $\frac{m_1 m_2 (1 + \mu_k)g}{(m_1 + m_2)}$ (D) $\frac{m_1 m_2 (1 - \mu_k)g}{(m_1 + m_2)}$

24. An electron moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the centre has magnitude

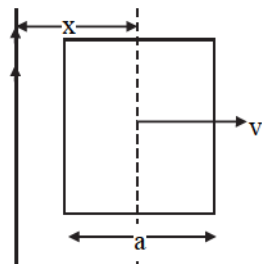
- (A) $\frac{\mu_0 n e}{2 \pi r}$ (B) Zero
 (C) $\frac{\mu_0 n^2 e}{r}$ (D) $\frac{\mu_0 n e}{2 r}$

25. Two spherical bodies of mass M and $5M$ and radii R and $2R$ are released in free space with initial separation between their centres equal to $12R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body before collision is

- (A) $2.5 R$ (B) $4.5 R$
 (C) $7.5 R$ (D) $1.5 R$

26. A conducting square frame of side ' a ' and a long straight wire carrying current I are located in the same plane as shown in the figure. The frame moves to the right with a constant velocity v . The e.m.f. induced in the frame will be proportional to

- (A) $\frac{1}{x^2}$
 (B) $\frac{1}{(2x - a)^2}$
 (C) $\frac{1}{(2x + a)^2}$
 (D) $\frac{1}{(2x - a)(2x + a)}$



27. A wind with speed 40 m/s blows parallel to the roof of a house. The area of the roof is 250 m^2 . Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind on the roof and the direction of the force will be ($\rho_{\text{air}} = 1.2 \text{ kg/m}^3$)

- (A) $4.8 \times 10^5 \text{ N}$, downwards.
 (B) $4.8 \times 10^5 \text{ N}$, upwards.
 (C) $2.4 \times 10^5 \text{ N}$, upwards.
 (D) $2.4 \times 10^5 \text{ N}$, downwards.

28. A radiation of energy ' E ' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is ($c = \text{velocity of light}$)

- (A) $\frac{E}{c}$ (B) $\frac{2E}{c}$ (C) $\frac{2E}{c^2}$ (D) $\frac{E}{c^2}$

29. A parallel plate air capacitor of capacitance C is connected to a cell of emf V and then disconnected from it. A dielectric slab of dielectric constant k , which can just fill the air gap of the capacitor, is now inserted in it. Which of the following is incorrect?

- (A) The potential difference between the plates decreases k times.
 (B) The energy stored in the capacitor decreases k times.
 (C) The change in energy stored is $\frac{1}{2} CV^2 \left(\frac{1}{k} - 1 \right)$.
 (D) The charge on the capacitor is not conserved.

30. A Carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J , the amount of energy absorbed from the reservoir at lower temperature is:

- (A) 1 J (B) 90 J
 (C) 99 J (D) 100 J

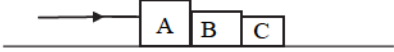
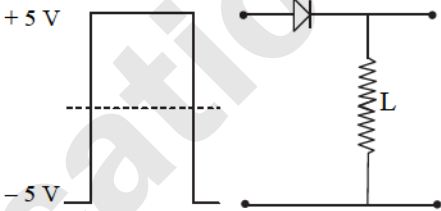
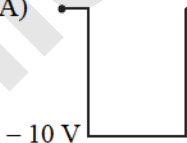
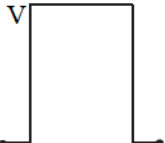
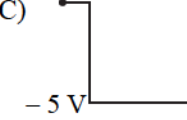
31. The refracting angle of a prism is A , and refractive index of the material of the prism is $\cot(A/2)$. The angle of minimum deviation is

- (A) $180^\circ - 3A$ (B) $180^\circ - 2A$
 (C) $90^\circ - A$ (D) $180^\circ + 2A$

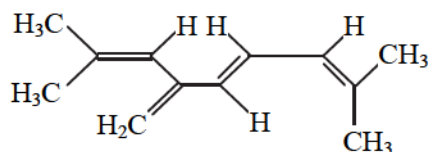
32. On observing light from three different stars P, Q and R, it was found that intensity of violet colour is maximum in the spectrum of P, the intensity of green colour is maximum in the spectrum of R and the intensity of red colour is maximum in the spectrum of Q. If T_P , T_Q and T_R are the respective absolute temperatures of P, Q and R then it can be concluded from the above observations that

- (A) $T_P > T_Q > T_R$ (B) $T_P > T_R > T_Q$
 (C) $T_P < T_R < T_Q$ (D) $T_P < T_Q < T_R$



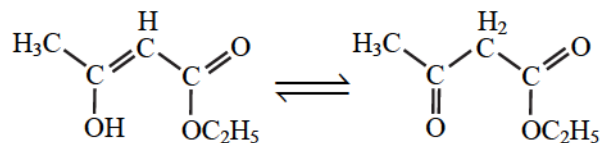
33. In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern?
 (A) 0.2 mm (B) 0.1 mm
 (C) 0.5 mm (D) 0.02 mm
34. The ratio of the specific heats $\frac{C_p}{C_v} = \gamma$ in terms of degrees of freedom (n) is given by
 (A) $(1 + \frac{1}{n})$ (B) $(1 + \frac{n}{3})$
 (C) $(1 + \frac{2}{n})$ (D) $(1 + \frac{n}{2})$
35. Three blocks A, B and C of masses 4 kg, 2 kg and 1 kg respectively, are in contact on a frictionless surface, as shown. If force of 14 N is applied on the 4 kg block, then the contact force between A and B is

 (A) 2 N (B) 6 N (C) 8 N (D) 18 N
36. The two ends of a metal rod are maintained at temperatures 100°C and 110°C. The rate of heat flow in the rod is found to be 4.0 J/s. If the ends are maintained at temperatures 200°C and 210°C, the rate of heat flow will be
 (A) 44.0 J/s (B) 16.8 J/s
 (C) 8.0 J/s (D) 4.0 J/s
37. A certain metallic surface is illuminated with monochromatic light of wavelength λ . The stopping potential for photo-electric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength for this surface for photoelectric effect is
 (A) 6λ (B) 4λ (C) $\frac{\lambda}{4}$ (D) $\frac{\lambda}{6}$
38. A resistance R draws power P when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes Z , the power drawn will be
 (A) $P(\frac{R}{Z})^2$ (B) $P\sqrt{\frac{R}{Z}}$
 (C) $P(\frac{R}{Z})$ (D) P
39. Consider 3rd orbit of He^+ (Helium), using non-relativistic approach, the speed of electron in this orbit will be [given $K = 9 \times 10^9$ constant, $Z = 2$ and h (Planck's Constant) = 6.6×10^{-34} J s]
 (A) 0.73×10^6 m/s (B) 3.0×10^8 m/s
 (C) 2.92×10^6 m/s (D) 1.46×10^6 m/s
40. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open on both ends. The length of organ pipe open at both the ends is
 (A) 80 cm (B) 100 cm
 (C) 120 cm (D) 140 cm
41. A block of mass 10 kg moving in x direction with a constant speed of 10 ms^{-1} , is subjected to a retarding force $F = 0.1x$ J/m during its travel from $x = 20$ m to 30 m. Its final KE will be
 (A) 475 J (B) 450 J
 (C) 275 J (D) 250 J
42. If in a p-n junction, a square input signal of 10 V is applied, as shown

 then the output across R_L will be
 (A)  (B) 
 (C)  (D) 
43. A potentiometer wire has length 4 m and resistance 8 Ω . The resistance that must be connected in series with the wire and an accumulator of e.m.f. 2 V, so as to get a potential gradient 1 mV per cm on the wire is
 (A) 32 Ω (B) 40 Ω
 (C) 44 Ω (D) 48 Ω
44. A particle of unit mass undergoes one-dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$, where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x , is given by
 (A) $-2n\beta^2 x^{-2n-1}$ (B) $-2n\beta^2 x^{-4n-1}$
 (C) $-2\beta^2 x^{-2n+1}$ (D) $-2n\beta^2 x^{-4n+1}$
45. The approximate depth of an ocean is 2700 m. The compressibility of water is $45.4 \times 10^{-11} \text{ Pa}^{-1}$ and density of water is 10^3 kg/m^3 . What fractional compression of water will be obtained at the bottom of the ocean?
 (A) 0.8×10^{-2} (B) 1.0×10^{-2}
 (C) 1.2×10^{-2} (D) 1.4×10^{-2}

46. The total number of π -bond electrons in the following structure is _____.



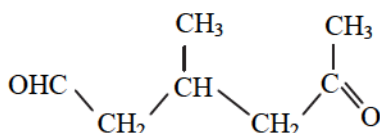
- (A) 4 (B) 8
(C) 12 (D) 16
47. Maximum bond angle at nitrogen is present in which of the following?
(A) NO_2 (B) NO_2^-
(C) NO_2^+ (D) NO_3^-
48. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12} , then at equilibrium the system will contain _____.
(A) all reactants
(B) mostly reactants
(C) mostly products
(D) similar amounts of reactants and products
49. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of $\text{Al}_2(\text{SO}_4)_3$ (if all are 100% ionised)?
(A) K_2SO_4 (B) $\text{K}_3[\text{Fe}(\text{CN})_6]$
(C) $\text{Al}(\text{NO}_3)_3$ (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$
50. Bithional is generally added to the soaps as an additive to function as a/an _____.
(A) softener (B) dryer
(C) buffering agent (D) antiseptic
51. "Metals are usually not found as nitrates in their ores." Out of the following two (I and II) reasons which is/are true for the above observation?
I. Metal nitrates are highly unstable.
II. Metal nitrates are highly soluble in water.
(A) I is false but II is true.
(B) I is true but II is false.
(C) I and II are true.
(D) I and II are false.
52. In Dumas method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is _____.
(A) 17.36 (B) 18.20
(C) 16.76 (D) 15.76
53. The correct bond order in the following species is _____.
(A) $\text{O}_2^+ < \text{O}_2^- < \text{O}_2^{2+}$ (B) $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2+}$
(C) $\text{O}_2^{2+} < \text{O}_2^+ < \text{O}_2^-$ (D) $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2^+$

54. The enolic form of ethyl acetoacetate as below has _____.



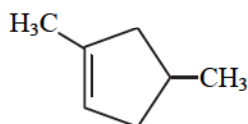
- (A) 18 sigma bonds and 2 pi bonds
(B) 16 sigma bonds and 1 pi bond
(C) 9 sigma bonds and 2 pi bonds
(D) 9 sigma bonds and 1 pi bond
55. In which of the following compounds, the C-Cl bond ionisation shall give most stable carbonium ion?
(A)
(B)
(C)
(D)
56. A given metal crystallizes out with a cubic structure having edge length of 361 pm. If there are four metal atoms in one unit cell, what is the radius of one atom?
(A) 40 pm (B) 127 pm
(C) 80 pm (D) 108 pm
57. The electrolytic reduction of nitrobenzene in strongly acidic medium produces _____.
(A) azobenzene (B) aniline
(C) p-aminophenol (D) azoxybenzene
58. When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is _____.
(A) zero
(B) first
(C) second
(D) more than zero but less than first
59. Which of the following statements is CORRECT for a reversible process in a state of equilibrium?
(A) $\Delta G^\circ = -2.30 RT \log K$
(B) $\Delta G^\circ = 2.30 RT \log K$
(C) $\Delta G = -2.30 RT \log K$
(D) $\Delta G = 2.30 RT \log K$

60. A single compound of the structure:

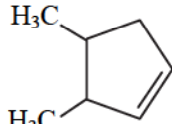


is obtainable from ozonolysis of which of the following cyclic compounds?

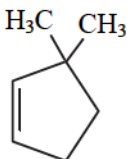
(A)



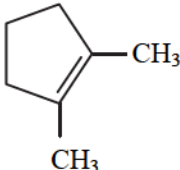
(B)



(C)



(D)



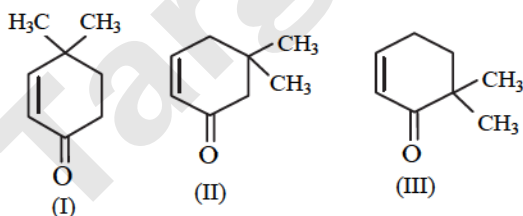
61. Which of the following processes does NOT involve oxidation of iron?

- (A) Formation of $\text{Fe}(\text{CO})_5$
 (B) Liberation of H_2 from steam by iron at high temperature
 (C) Rusting of iron sheets.
 (D) Decolourisation of blue CuSO_4 solution by iron.

62. The K_{sp} of Ag_2CrO_4 , AgCl , AgBr and AgI are respectively, 1.1×10^{-12} , 1.8×10^{-10} , 5.0×10^{-13} and 8.3×10^{-17} . Which one of the following salts will precipitate last if AgNO_3 solution is added to the solution containing equal moles of NaCl , NaBr , NaI and Na_2CrO_4 ?

- (A) AgI (B) AgCl
 (C) AgBr (D) Ag_2CrO_4

63. Given:



Which of the given compounds can exhibit tautomerism?

- (A) II and III (B) I, II and III
 (C) I and II (D) I and III

64. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture?

- (A) 16 : 1 (B) 2 : 1
 (C) 1 : 4 (D) 4 : 1

65. Which of these statements about $[\text{Co}(\text{CN})_6]^{3-}$ is CORRECT?

- (A) It has no unpaired electrons and has low-spin configuration.
 (B) It has four unpaired electrons and has low-spin configuration.
 (C) It has four unpaired electrons and has high-spin configuration.
 (D) It has no unpaired electrons and has high-spin configuration.

66. Solubility of the alkaline earth metal sulphates in water decreases in the sequence _____.

- (A) $\text{Mg} > \text{Ca} > \text{Sr} > \text{Ba}$
 (B) $\text{Ca} > \text{Sr} > \text{Ba} > \text{Mg}$
 (C) $\text{Sr} > \text{Ca} > \text{Mg} > \text{Ba}$
 (D) $\text{Ba} > \text{Mg} > \text{Sr} > \text{Ca}$

67. Which property of colloidal solution is independent of charge on the colloidal particles?

- (A) Coagulation
 (B) Electrophoresis
 (C) Electroosmosis
 (D) Tyndall effect

68. A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as _____.

- (A) fuel cell (B) electrolytic cell
 (C) dynamo (D) Ni-Cd cell

69. Which of the following pairs of ions are isoelectronic and isostructural?

- (A) SO_3^{2-} , NO_3^- (B) ClO_3^- , SO_3^{2-}
 (C) CO_3^{2-} , SO_3^{2-} (D) ClO_3^- , CO_3^{2-}

70. Which one is NOT equal to zero for an ideal solution?

- (A) $\Delta_{\text{mix}}H$
 (B) $\Delta_{\text{mix}}S$
 (C) $\Delta_{\text{mix}}V$
 (D) $\Delta P = P_{\text{Observed}} - P_{\text{Raoult}}$

71. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers)

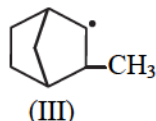
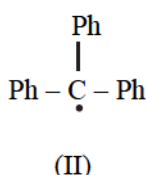
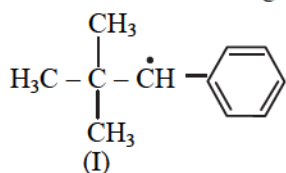
- (A) Ti (22) and Zr (40)
 (B) Zr (40) and Nb (41)
 (C) Zr (40) and Hf (72)
 (D) Zr (40) and Ta (73)

72. The number of d-electrons in Fe^{2+} ($Z = 26$) is NOT equal to the number of electrons in which one of the following?

- (A) s-electrons in Mg ($Z = 12$)
 (B) p-electrons in Cl ($Z = 17$)
 (C) d-electrons in Fe ($Z = 26$)
 (D) p-electrons in Ne ($Z = 10$)

73. Biodegradable polymer which can be produced from glycine and aminocaproic acid is _____.
 (A) nylon-2-nylon 6 (B) PHBV
 (C) buna-N (D) nylon-6,6

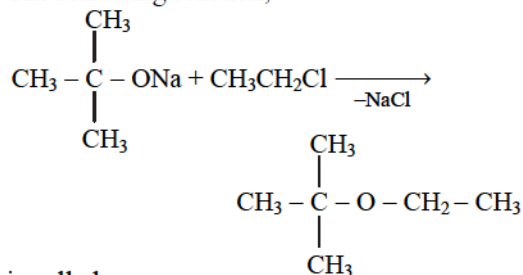
74. Consider the following compounds:



Hyperconjugation occurs in _____.

- (A) I only (B) II only
 (C) III only (D) I and III
75. The angular momentum of electron in 'd' orbital is equal to _____.
 (A) $2\sqrt{3}h$ (B) $0h$
 (C) $\sqrt{6}h$ (D) $\sqrt{2}h$
76. The activation energy of a reaction can be determined from the slope of which of the following graphs?
 (A) $\frac{\ln k}{T}$ vs. T (B) $\ln k$ vs. $\frac{1}{T}$
 (C) $\frac{1}{\ln k}$ vs. $\frac{1}{T}$ (D) $\ln k$ vs. T
77. The function of 'sodium pump' is a biological process operating in each and every cell of all animals. Which of the following biologically important ions is also a constituent of the pump?
 (A) Ca^{2+} (B) Mg^{2+}
 (C) K^+ (D) Fe^{2+}
78. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is TRUE in this case?
 (A) X is undergoing dissociation in water.
 (B) Molecular mass of X is greater than the molecular mass of Y.
 (C) Molecular mass of X is less than the molecular mass of Y.
 (D) Y is undergoing dissociation in water while X undergoes no change.

79. The following reaction,



is called _____.

- (A) Williamson synthesis
 (B) Williamson continuous etherification process
 (C) Etard reaction
 (D) Gatterman - Koch reaction
80. Cobalt(III) chloride forms several octahedral complexes with ammonia. Which of the following will NOT give test for chloride ions with silver nitrate at 25°C ?
 (A) $\text{CoCl}_3 \cdot 3\text{NH}_3$ (B) $\text{CoCl}_3 \cdot 4\text{NH}_3$
 (C) $\text{CoCl}_3 \cdot 5\text{NH}_3$ (D) $\text{CoCl}_3 \cdot 6\text{NH}_3$
81. Which of the following is the most CORRECT electron displacement for a nucleophilic reaction to take place?
 (A) $\text{H}_3\text{C} \rightarrow \text{C}=\text{C}-\text{CH}_2-\text{Cl}$
 (B) $\text{H}_3\text{C} \leftarrow \text{C}=\text{C}-\text{CH}_2-\text{Cl}$
 (C) $\text{H}_3\text{C} \rightarrow \text{C}=\text{C}-\text{CH}_2-\text{Cl}$
 (D) $\text{H}_3\text{C} \rightarrow \text{C}=\text{C}-\text{CH}_2-\text{Cl}$
82. Nitrogen dioxide and sulphur dioxide have some properties in common. Which property is shown by one of these compounds, but NOT by the other?
 (A) Is a reducing agent
 (B) Is soluble in water
 (C) Is used as a food-preservative
 (D) Forms 'acid-rain'
83. The species Ar, K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase?
 (A) $\text{Ar} < \text{K}^+ < \text{Ca}^{2+}$ (B) $\text{Ca}^{2+} < \text{Ar} < \text{K}^+$
 (C) $\text{Ca}^{2+} < \text{K}^+ < \text{Ar}$ (D) $\text{K}^+ < \text{Ar} < \text{Ca}^{2+}$
84. Which of the following options represents the CORRECT bond order?
 (A) $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$ (B) $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$
 (C) $\text{O}_2^- > \text{O}_2 < \text{O}_2^+$ (D) $\text{O}_2^- < \text{O}_2 > \text{O}_2^+$
85. The reaction of $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_3$ with HBr produces _____.
 (A) $\text{C}_6\text{H}_5\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$
 (B) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$
 (C) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
 (D) $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_2\text{Br}$



86. An organic compound 'X' having molecular formula $C_5H_{10}O$ yields phenyl hydrazone and gives negative response to the iodoform test and Tollen's test. It produces n-pentane on reduction. 'X' could be _____.

(A) pentanal (B) pentan-2-one
(C) pentan-3-one (D) n-amyl alcohol

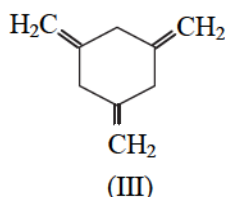
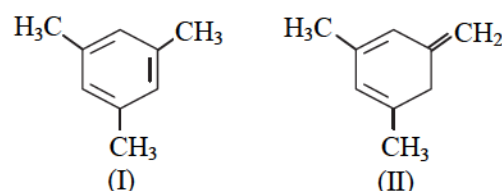
87. Magnetic moment 2.84 B.M. is given by _____. (At. Numbers: Ni = 28, Ti = 22, Cr = 24, Co = 27)

(A) Ni^{2+} (B) Ti^{3+}
(C) Cr^{2+} (D) Co^{2+}

88. Which of the following species contains equal number of σ - and π -bonds?

(A) HCO_3^- (B) XeO_4
(C) $(CN)_2$ (D) $CH_2(CN)_2$

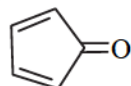
89. Given:



The enthalpy of hydrogenation of these compounds will be in the order as _____.

(A) $I > II > III$ (B) $III > II > I$
(C) $II > III > I$ (D) $II > I > III$

90. Treatment of cyclopentanone with methyl lithium gives which of the following species?



Cyclopentanone

(A) Cyclopentanonyl radical
(B) Cyclopentanonyl biradical
(C) Cyclopentanonyl anion
(D) Cyclopentanonyl cation

91. Which one of the following matches is CORRECT?

(A)	<i>Alternaria</i>	Sexual reproduction absent	Deuteromycetes
(B)	<i>Mucor</i>	Reproduction by conjugation	Ascomycetes
(C)	<i>Agaricus</i>	Parasitic fungus	Phycomycetes
(D)	<i>Trichoderma</i>	Aseptate mycelium	Basidiomycetes

92. Which one of the following statements is NOT true?

(A) Pollen grains are rich in nutrients, and they are used in the form of tablets and syrups.
(B) Pollen grains of some plants cause severe allergies and bronchial afflictions in some people.
(C) The flowers pollinated by flies and bats secrete foul odour to attract them.
(D) Honey is made by bees by digesting pollen collected from flowers.

93. Which of the following animals is NOT viviparous?

(A) Elephant (B) Platypus
(C) Whale (D) Flying fox (Bat)

94. Which is the most common mechanism of genetic variation in the population of sexually reproducing organism?

(A) Chromosomal aberrations
(B) Genetic drift
(C) Recombination
(D) Transduction

95. Which one of the following statements is WRONG?

(A) *Chlorella* and *Spirulina* are used as space food.
(B) Mannitol is stored food in Rhodophyceae.
(C) Algin and carrageenan are products of algae.
(D) Agar-agar is obtained from *Gelidium* and *Gracilaria*.

96. Capacitation refers to changes in the

(A) ovum after fertilisation
(B) sperm after fertilisation
(C) sperm before fertilisation
(D) ovum before fertilization

97. Leaves become modified into spines in

(A) onion (B) silk cotton
(C) *Opuntia* (D) pea

98. Which of the following is NOT a sexually transmitted disease?

(A) Syphilis
(B) Acquired Immuno Deficiency Syndrome (AIDS)
(C) Trichomoniasis
(D) Encephalitis

99. Select the CORRECT match in the following pairs.

(A) Smooth ER – Oxidation of phospholipids
(B) Smooth ER – Synthesis of lipids
(C) Rough ER – Synthesis of glycogen
(D) Rough ER – Oxidation of fatty acids



100. Which of the following had the smallest brain capacity?
 (A) *Homo erectus*
 (B) *Homo sapiens*
 (C) *Homo neanderthalensis*
 (D) *Homo habilis*
101. A major characteristic of the monocot root is the presence of
 (A) scattered vascular bundles
 (B) vasculature without cambium
 (C) cambium sandwiched between phloem and xylem along the radius
 (D) open vascular bundles
102. HIV that causes AIDS, first starts destroying
 (A) B-Lymphocytes
 (B) leucocytes
 (C) helper T-Lymphocytes
 (D) thrombocytes
103. A somatic cell that has just completed the S phase of its cell cycle, as compared to gamete of the same species, has
 (A) twice the number of chromosomes and twice the amount of DNA.
 (B) same number of chromosomes but twice the amount of DNA.
 (C) twice the number of chromosomes and four times the amount of DNA.
 (D) four times the number of chromosomes and twice the amount of DNA.
104. A man with blood group 'A' marries a woman with blood group 'B'. What are all the possible blood groups of their offsprings?
 (A) A, B and AB only (B) A, B, AB and O
 (C) O only (D) A and B only
105. In a ring girdled plant:
 (A) the root dies first
 (B) the shoot and root die together
 (C) neither root nor shoot will die
 (D) the shoot dies first
106. Which of the following enhances or induces fusion of protoplasts?
 (A) IAA and kinetin
 (B) IAA and gibberellins
 (C) Sodium chloride and potassium chloride
 (D) Polyethylene glycol and sodium nitrate
107. Read the following five statements (i to v) and select the option with all correct statements:
 i. Mosses and lichens are the first organisms to colonise a bare rock.
 ii. *Selaginella* is a homosporous pteridophyte.
 iii. Coralloid roots in *Cycas* have VAM.
 iv. Main plant body in bryophytes is gametophytic, whereas in pteridophytes it is sporophytic.
 v. In gymnosperms, male and female gametophytes are present within sporangia located on sporophyte.
- (A) (ii), (iii) and (iv) (B) (i), (iv) and (v)
 (C) (ii), (iii) and (v) (D) (i), (iii) and (iv)
108. In Sea Urchin DNA, which is double stranded 17% of the bases were shown to be cytosine. The percentages of the other three bases expected to be present in this DNA are
 (A) G 17%, A 16.5%, T 32.5%
 (B) G 17%, A 33%, T 33%
 (C) G 8.5%, A 50%, T 24.5%
 (D) G 34%, A 24.5%, T 24.5%
109. Male gametes are flagellated in :
 (A) *Anabaena* (B) *Ectocarpus*
 (C) *Spirogyra* (D) *Polysiphonia*
110. How many pairs of contrasting characters in pea plants were studied by Mendel in his experiments?
 (A) Six (B) Eight
 (C) Seven (D) Five
111. Perigynous flowers are found in
 (A) Cucumber (B) China rose
 (C) Rose (D) Guava
112. Which of these is NOT an important component of initiation of parturition in humans?
 (A) Release of oxytocin
 (B) Release of prolactin
 (C) Increase in estrogen and progesterone ratio
 (D) Synthesis of prostaglandins
113. True nucleus is absent in :-
 (A) *Mucor* (B) *Vaucheria*
 (C) *Volvox* (D) *Anabaena*
114. Gene regulation governing lactose operon of *E. coli* that involves the *lac I* gene product is
 (A) positive and inducible because it can be induced by lactose.
 (B) negative and inducible because repressor protein prevents transcription.
 (C) negative and repressible because repressor protein prevents transcription.
 (D) feedback inhibition because excess of β -galactosidase can switch off transcription.
115. Select the correct option.

	Column I		Column II
i.	Synapsis aligns homologous chromosomes	a.	Anaphase II
ii.	Synthesis of RNA and protein	b.	Zygotene
iii.	Action of enzyme recombinase	c.	G ₂ phase
iv.	Centromeres do not separate but chromatids move towards opposite poles	d.	Anaphase I
		e.	Pachytene

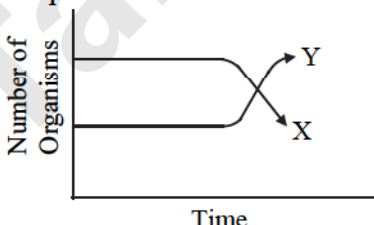


- (A) i-b, ii-a, iii-c, iv-d
(B) i-b, ii-c, iii-e, iv-d
(C) i-a, ii-b, iii-e, iv-d
(D) i-b, ii-c, iii-d, iv-e
116. Rachel Carson's famous book "Silent Spring" is related to
(A) Noise pollution
(B) Population explosion
(C) Ecosystem management
(D) Pesticide pollution
117. In which of the following gametophyte is not independent free living ?
(A) *Marchantia* (B) *Pteris*
(C) *Pinus* (D) *Fumaria*
118. Secondary succession takes place on/in
(A) Degraded forest
(B) Newly created pond
(C) Newly cooled lava
(D) Bare rock
119. $\oplus K_{(5)} \overline{C}_{(5)} A_5 \underline{G}_{(2)}$ is the floral formula of
(A) *Sesbania* (B) *Petunia*
(C) *Brassica* (D) *Allium*
120. Most animals are tree dwellers in a
(A) temperate deciduous forest
(B) tropical rainforest
(C) coniferous forest
(D) thorn woodland
121. Vascular bundles in monocotyledons are considered closed because
(A) cambium is absent
(B) there are no vessels with perforations
(C) xylem is surrounded all around by phloem
(D) a bundle sheath surrounds each bundle
122. Cryopreservation of gametes of threatened species in viable and fertile condition can be referred to as
(A) Advanced *ex-situ* conservation of biodiversity
(B) *In situ* conservation by sacred groves
(C) *In situ* cryoconservation of biodiversity
(D) *In situ* conservation of biodiversity
123. Which of the following characteristics is mainly responsible for diversification of insects on land?
(A) Bilateral symmetry
(B) Exoskeleton
(C) Eyes
(D) Segmentation
124. Which one of the following is NOT an inclusion body found in prokaryotes?
(A) Phosphate granule
(B) Cyanophycean granule
(C) Glycogen granule
(D) Polysome
125. Which one of the following may require pollinators, but is genetically similar to autogamy?
(A) Geitonogamy (B) Xenogamy
(C) Apogamy (D) Cleistogamy
126. Minerals known to be required in large amounts for plant growth include
(A) potassium, phosphorus, selenium, boron
(B) magnesium, sulphur, iron, zinc
(C) phosphorus, potassium, sulphur, calcium
(D) calcium, magnesium, manganese, copper
127. Match each disease with its correct type of vaccine.
- | | Column I | | Column II |
|------|----------------|----|-------------------|
| i. | Tuberculosis | a. | Harmless virus |
| ii. | Whooping cough | b. | Inactivated toxin |
| iii. | Diphtheria | c. | Killed bacteria |
| iv. | Polio | d. | Harmless bacteria |
- (A) i-(d), ii-(c), iii-(b), iv-(a)
(B) i-(a), ii-(b), iii-(d), iv-(c)
(C) i-(b), ii-(a), iii-(c), iv-(d)
(D) i-(c), ii-(b), iii-(d), iv-(a)
128. Transmission tissue is characteristic feature of
(A) Solid style (B) Dry stigma
(C) Wet stigma (D) Hollow style
129. Multiple alleles are present
(A) at the same locus of the chromosome
(B) on non-sister chromatids
(C) on different chromosomes
(D) at different loci on the same chromosome
130. The terga, sterna and pleura of cockroach body are joined by
(A) cementing glue
(B) muscular tissue
(C) arthrodial membrane
(D) cartilage
131. Which of the following viruses is NOT transferred through semen of an infected male?
(A) Chikungunya virus
(B) Ebola virus
(C) Hepatitis B virus
(D) Human immunodeficiency virus
132. Nuclear envelope is a derivative of
(A) smooth endoplasmic reticulum
(B) membrane of Golgi complex
(C) microtubules
(D) rough endoplasmic reticulum
133. Which of the following are the important floral rewards to the animal pollinators?
(A) Colour and large size of flower
(B) Nectar and pollen grains
(C) Floral fragrance and calcium crystals
(D) Protein pellicle and stigmatic exudates



134. Which of the following endoparasites of humans does show viviparity ?
 (A) *Enterobius vermicularis*
 (B) *Trichinella spiralis*
 (C) *Ascaris lumbricoides*
 (D) *Ancylostoma duodenale*
135. The movement of a gene from one linkage group to another is called
 (A) Duplication (B) Translocation
 (C) Crossing over (D) Inversion
136. Gastric juice of infants contains
 (A) maltase, pepsinogen, rennin
 (B) nuclease, pepsinogen, lipase
 (C) pepsinogen, lipase, rennin
 (D) amylase, rennin, pepsinogen
137. Which one gives the most valid and recent explanation for stomatal movement?
 (A) Starch hydrolysis
 (B) Guard cell photosynthesis
 (C) Transpiration
 (D) Potassium influx and efflux
138. The active form of *Entamoeba histolytica* feeds upon
 (A) food in intestine
 (B) blood only
 (C) erythrocytes, mucosa and submucosa of colon
 (D) mucosa and submucosa of colon only
139. Which of the following represents the correct combination without any exception?
- | | Characteristics | Class |
|-----|--|----------------|
| (1) | Mouth ventral, gills without operculum; skin with placoid scales; persistent notochord | Chondrichthyes |
| (2) | Sucking and circular mouth; jaws absent, integument without scales; paired appendages | Cyclostomata |
| (3) | Body covered with feathers; skin moist and glandular; fore-limbs form wings; lungs with air sacs | Aves |
| (4) | Mammary gland; hair on body; pinnae; two pairs of Limbs | Mammalia |
140. The crops engineered for glyphosate are resistant/tolerant to
 (A) insects (B) herbicides
 (C) fungi (D) bacteria
141. Glenoid cavity articulates
 (A) clavicle with scapula
 (B) humerus with scapula
 (C) clavicle with acromion
 (D) scapula with acromion
142. Transpiration and root pressure cause water to rise in plants by
 (A) pulling it upward
 (B) pulling and pushing it, respectively
 (C) pushing it upward
 (D) pushing and pulling it, respectively
143. The mass of living material at a trophic level at a particular time is called :
 (A) Standing state
 (B) Net primary productivity
 (C) Standing crop
 (D) Gross primary productivity
144. Keel is the characteristic feature of flower of
 (A) *Aloe* (B) tomato
 (C) tulip (D) *Indigofera*
145. Which of the following cells during gametogenesis is normally diploid?
 (A) Primary polar body
 (B) Spermatid
 (C) Spermatogonia
 (D) Secondary polar body
146. Root pressure develops due to
 (A) passive absorption
 (B) active absorption
 (C) increase in transpiration
 (D) low osmotic potential in soil
147. A population will not exist in Hardy-Weinberg equilibrium if
 (A) individuals mate selectively
 (B) there are no mutations
 (C) there is no migration
 (D) the population is large
148. The chromosomes in which centromere is situated close to one end are
 (A) telocentric (B) sub-metacentric
 (C) metacentric (D) acrocentric
149. A technique of micropropagation is :-
 (A) Somatic embryogenesis
 (B) Protoplast fusion
 (C) Embryo rescue
 (D) somatic hybridization
150. A gymnast is able to balance his body upside down even in the total darkness because of
 (A) cochlea
 (B) vestibular apparatus
 (C) tectorial membrane
 (D) organ of Corti
151. Cytochromes are found in
 (A) matrix of mitochondria
 (B) outer wall of mitochondria
 (C) cristae of mitochondria
 (D) lysosomes



152. In which of the following both pairs have correct combination?
(A) *In situ* conservation: Cryopreservation
Ex situ conservation: Wildlife Sanctuary
(B) *In situ* conservation: Seed Bank
Ex situ conservation: National Park
(C) *In situ* conservation: Tissue culture
Ex situ conservation: Sacred groves
(D) *In situ* conservation: National Park
Ex situ conservation: Botanical Garden
153. The structures that are formed by stacking of organized flattened membranous sacs in the chloroplasts are
(A) cristae (B) grana
(C) stroma lamellae (D) stroma
154. When you hold your breath, which of the following gas changes in blood would first lead to the urge to breathe?
(A) Falling CO_2 concentration
(B) Rising CO_2 and falling O_2 concentration
(C) Falling O_2 concentration
(D) Rising CO_2 concentration
155. The hilum is a scar on the
(A) Seed, where funicle was attached
(B) Fruit, where it was attached to pedicel
(C) Fruit, where style was present
(D) Seed, where micropyle was present
156. What causes a green plant exposed to the light, on only one side, to bend toward the source of light as it grows?
(A) Light stimulates plant cells on the lighted side to grow faster.
(B) Auxin accumulates on the shaded side, stimulating greater cell elongation there.
(C) Green plants need light to perform photosynthesis.
(D) Green plants seek light because they are phototropic.
157. The following graph depicts changes in two populations (X and Y) of herbivores in a grassy field. X possible reason for these changes is that
- 
- (A) both plant populations in this habitat decreased
(B) population Y competed more successfully for food than population X
(C) population X produced more offspring than population Y
(D) population X consumed the members of population Y
158. Which of the following statement is NOT correct?
(A) Brunner's glands are present in the submucosa of stomach and secrete pepsinogen.
(B) Goblet cells are present in the mucosa of intestine and secrete mucus.
(C) Oxyntic cells are present in the mucosa of stomach and secrete HCl.
(D) Acini are present in the pancreas and secrete carboxypeptidase.
159. Alleles are
(A) true breeding homozygotes
(B) different molecular forms of a gene
(C) heterozygotes
(D) different phenotype
160. Typical growth curve in plants is
(A) linear (B) stair-steps shaped
(C) parabolic (D) sigmoid
161. High value of BOD (Biochemical Oxygen Demand) indicates that :-
(A) Water is highly polluted
(B) Water is less polluted
(C) Consumption of organic matter in the water is higher by the microbes
(D) Water is pure
162. Which one of the following is CORRECT?
(A) Plasma = Blood – Lymphocytes
(B) Serum = Blood + Fibrinogen
(C) Lymph = Plasma + RBC + WBC
(D) Blood = Plasma + RBC + WBC + Platelets
163. Hysterectomy is surgical removal of
(A) vas deferens (B) mammary glands
(C) uterus (D) prostate gland
164. DNA is NOT present in
(A) Ribosomes (B) Nucleus
(C) Mitochondria (D) Chloroplast
165. An abnormal human baby with 'XXX' sex chromosomes was born due to
(A) fusion of two ova and one sperm
(B) fusion of two sperms and one ovum
(C) formation of abnormal sperms in the father
(D) formation of abnormal ova in the mother
166. Blood pressure in the mammalian aorta is maximum during
(A) diastole of the right ventricle
(B) systole of the left ventricle
(C) diastole of the right atrium
(D) systole of the left atrium
167. In Bt cotton, the Bt toxin present in plant tissue as protoxin is converted into active toxin due to
(A) acidic pH of the insect gut
(B) action of gut microorganisms
(C) presence of conversion factors in insect gut
(D) alkaline pH of the insect gut



168. Removal of proximal convoluted tubule from the nephron will result in
(A) no change in quality and quantity of urine
(B) no urine formation
(C) more diluted urine
(D) more concentrated urine
169. Which body of the Government of India regulates GM research and safety of introducing GM organisms for public services?
(A) Bio-safety committee
(B) India Council of Agricultural Research
(C) Genetic Engineering Approval Committee
(D) Research Committee on Genetic Manipulation
170. A chemical signal that has both endocrine and neural roles is
(A) melatonin (B) calcitonin
(C) epinephrine (D) cortisol
171. Which one of the following statements is INCORRECT?
(A) A competitive inhibitor reacts reversibly with the enzyme to form an enzyme inhibitor complex.
(B) In competitive inhibition, the inhibitor molecule is not chemically changed by the enzyme.
(C) The competitive inhibitor does not affect the rate of breakdown of the enzyme-substrate complex.
(D) The presence of the competitive inhibitor decreases the K_m of the enzyme for the substrate.
172. The UN Conference of Parties on climate change in the year 2011 was held in
(A) South Africa (B) Peru
(C) Qatar (D) Poland
173. Sliding filament theory can be best explained as
(A) when myofilaments slide pass each other actin filaments shorten, while myosin filament do not shorten
(B) actin and myosin filaments shorten and slide pass each other
(C) actin and myosin filaments do not shorten but rather slide pass each other
(D) when myofilaments slide pass each other, myosin filaments shorten while actin filaments do not shorten
174. In an ecosystem, the rate of production of organic matter during photosynthesis is termed as
(A) Gross primary productivity
(B) Secondary productivity
(C) Net productivity
(D) Net primary productivity
175. Which of the following does not favour the formation of large quantities of dilute urine?
(A) Renin
(B) Atrial-natriuretic factor
(C) Alcohol
(D) Caffeine
176. The guts of cow and buffalo possess
(A) Methanogens (B) Cyanobacteria
(C) *Fucus* sp. (D) *Chlorella* sp.
177. Which of the following regions of the brain is INCORRECTLY paired with its function?
(A) Medulla oblongata – Homeostatic control
(B) Cerebellum – Language comprehension
(C) Corpus callosum – Communication between the left and the right cerebral cortices
(D) Cerebrum – Calculation and contemplation
178. Vertical distribution of different species occupying different levels in a biotic community is known as
(A) Stratification (B) Zonation
(C) Pyramid (D) Divergence
179. In ginger vegetative propagation occurs through
(A) Rhizome (B) Offsets
(C) Bulbils (D) Runners
180. Erythropoiesis starts in
(A) spleen (B) red bone marrow
(C) kidney (D) liver